

SUBSTATION CONDITION

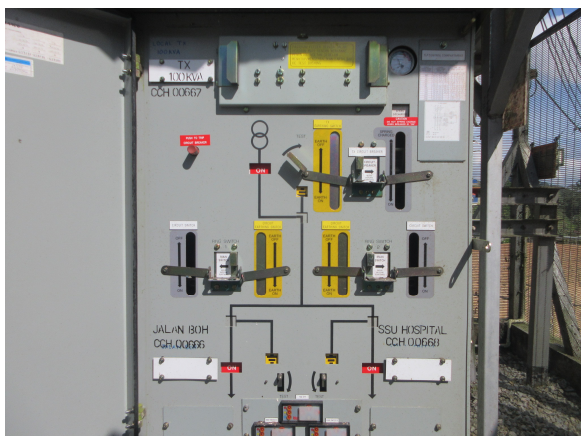
SUBSTATION OVERVIEW



SIGNBOARD



SWITCHGEAR



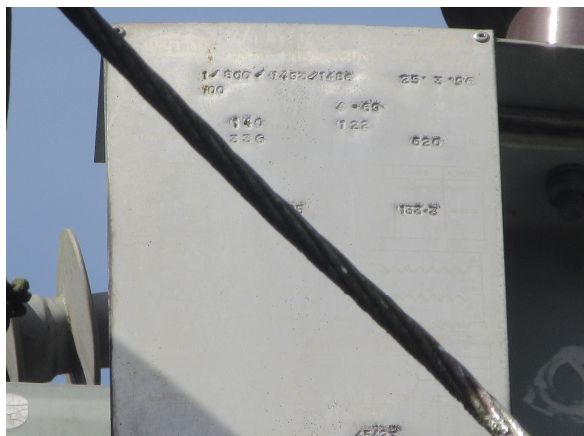
SWITCHGEAR NAMEPLATE



TRANSFORMER



TRANSFORMER NAMEPLATE



SUBSTATION CONDITION

EFI



SF6 INDICATOR



TRANSFORMER OIL LEVEL INDICATOR



VISUAL DEFECT



CPR POSTER OLD VERSION – SWG BAY

NORMAL/DEFECT STICKER

NORMAL STICKER	
CONDITION ASSESSMENT  TENAGA NASIONAL <i>Enter Right</i> DIAGNOSTIC UNIT SUBSTATION 	
Cycle/Year :	1/2026
Testing Date :	01-06-2026
Defect :	—
Company/Team :	EET ENGINEERING
	TEV U/S IR

151

PAT+
RMU

STATE / AREA:		PAHANG / CAMELON HIGHLAND		DISTRIBUTION SUBSTATION (PCR) CMA TEST SHEET		DATE		16/2026		FUNCTIONAL LOCATION:		CUM/PCB/J00002													
PE NAME:		JALAN ROBINSON		TIME IN / OUT		4:02 P.M		4:17 P.M		BACKGROUND TEMP:		23.2 °C													
V.I REF. NO				TEV BACK GROUND / HUMIDITY		1		1																	
DETAIL																									
NO.	FDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE/SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLIV)	COMP	Tmin (°C)	Tmax (°C)	AT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE / CYCLE	TYPE CHAR.	REMARK				
1	CLH 00666	JALAN BOLA	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	31.9	32.1	0.2	32.0	6.7		0	2	-	5	0	-					
2	CLH 00667	TX 100 KVA	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	31.4	31.7	0.3	31.5	8.9		0	3	-	4	0	-					
3	CLH 00668	SSU HOSPITAL	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	38.4	38.9	0.5	38.6	10.11	SOLAR LOADING	0	4	-	4	0	-					
4			TRIP / CLOSE		XLPE PILC N/A NACC			CABLE																	
OVERVIEW								Bottom		39.0		1		SOLAR LOADING						01					
LOCAL TRANSFORMER																									
DETAILS				THERMOGRAPHY				ULTRASOUND				DETAILS				THERMOGRAPHY				ULTRASOUND					
TX ID	SERIAL NO	CABLE TYPE/SIZE	REMARK	COMP.	Tmin (°C)	Tmax (°C)	AT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/SIZE	COMP.	Tmin (°C)	Tmax (°C)	AT	FILE	Avg (°C)	FILE	dB	TYPE CHAR.	REMARKS	
TX1	11800 / 1452 / 1486	XLPE/PILC		HT CABLE	32.0	32.3	0.3	32.2	17.18	0	-	TX3		XLPE/PILC	HT CABLE	1									
		HT BUSHING	30.9	31.1	0.2	31.0	16	0	-		XLPE/PILC		HT BUSHING												
		LV CABLE	27.2	27.5	0.3	27.3	15				XLPE/PILC		LV CABLE												
		LV BUSHING	28.8	29.2	0.4	28.9	14				XLPE/PILC		LV BUSHING												
TX2		XLPE/PILC		HT CABLE	36.3	41.7	5.4	38.5	13		-	TX4		XLPE/PILC	HT CABLE										
		HT BUSHING									XLPE/PILC		HT BUSHING												
		LV CABLE									XLPE/PILC		LV CABLE												
		LV BUSHING									XLPE/PILC		LV BUSHING												
				BODY											BODY										
LV DISTRIBUTION BOARD / FEEDER PILLAR / OTHERS																									
1	LVDB / FP 1	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10											
	CABLE TYPE / SIZE	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC											
2	LVDB / FP 2	INC. 1	INC. 2	INC. 3	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10											
	CABLE TYPE / SIZE	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC											
NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin (°C)	Tmax (°C)	AT	FILE	Avg (°C)	REMARK							LVDB / FP 1 3-SLOTTED / DIN TYPE										
	INC 1														Manufacturer:										
	INC 2														Serial No.:										
	F1														Rating:										
	F2														1600A / 1000A / 800A										
	F3														LVDB / FP 2 3-SLOTTED / DIN TYPE										
	F4														Manufacturer:										
	F5														Serial No.:										
	F6														Rating:										
	F7														1600A / 1000A / 800A										
	F8																								
	Battery Bank																								
	Black Box 1																								
	Black Box 2																								
	Lightning Arrester																								
	Dropout Fuse																								
	Auto Recloser																								
	LBS																								

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MOHD AZZAD BIN OTHMAN (10085230)
 JURUTEKNIK TINGKATAN BIASA
 (DIAGNOSTIC SUBSTATION PAHANG)
 SBU MAINTENANCE EAST
 DISTRIBUTION NETWORK DIVISION, TNB

151

PAT+RMU

CPR (O.V, Sur Bag)



VISUAL INSPECTION CHECKLIST FOR 11KV SUBSTATION

DATE:

P/E Name (ERMS):

P/E Name (SITE):

GPS Coordinate:

PCE Type:

PCE Area

Switchgear Type 1:

Manufacturer/Modal:

Switchgear Manufactured Date:

Switchgear Type 2:

Manufacturer/Modal:

Switchgear Manufactured Date:

No. of Transformer:

Tx 1:

Tx 2:

Tx 3:

Tx 4:

KEJANGGALAN

1. Signboard:

2. Earth connection:

3. EFI Functioning:

4. Heater Functioning:

5. Labelling @Switchgear:

6. Labelling @ LVDB:

7. Gas Pressure:

8. Tx Guard:

9. Tx Oil Level (conservator tank):

10. Tx Oil Leak:

11. Tx Bushing:

12. Top Demand Indicator (@LVDB):

13. Link/fuse Holder:

14. LV Termination:

15. Rubber Mat:

16. Trenching:

17. Battery Voltage:

18. Lighting:

19. Roof Condition:

20. Fire Fighting

21. Fence/door condition:

22. Cleanliness/Bushes:

23. Vandalism

24. Other Findings

1/6/2026

JALAN ROBINSON

JALAN ROBINSON

4.453222, 101.391489

INDOOR

NORMAL

RMU OIL

JAMCO / GRI

2017

RMU OIL

RMU SF6

1

Type HS

Type

Type

Type

Type

KVA 100

Construction Yr: 1996

Manufacturer: MTR

S/N: T1800/1452/1486

Construction Yr:

Manufacturer:

Construction Yr:

Manufacturer:

Construction Yr:

Manufacturer:

Construction Yr:

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Manufacturer:

F/L NO (ERMS):

F/L NO (SITE):

CCAL / PCE / 100002

CCAL / PCE / 100002

ATTACH BUILDING

COMPACT

COASTAL

MRMU

VCB

OCB

RMU SF6 / OILS/N:

RMU SF6 / OILS/N:

Switchgear Rating:

Switchgear Rating:

Construction Yr:

Manufacturer:

Construction Yr:

Manufacturer:

Construction Yr:

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PAT

COASTAL

MRMU

VCB

OCB

RMU SF6 / OILS/N:

RMU SF6 / OILS/N:

Switchgear Rating:

Switchgear Rating:

Construction Yr:

Manufacturer:

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RMU SF6 / OILS/N: B 21729024

OCB

RMU SF6 / OILS/N:

RMU SF6 / OILS/N:

Switchgear Rating:

Switchgear Rating:

Construction Yr:

Manufacturer:

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TNB SUPERVISOR/ ENGINEER (NAME & SIGNATURE):

DISTRIBUTION NETWORK DIVISION, TNB